Name:		Index Number:		Class:	

DUNMAN HIGH SCHOOL
Preliminary Exam
Year 6

H2 Biology

9744/04

Paper 4 Practical

24 Aug 2018
2 hour 30 minutes

Candidates answer on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Write your name, class index number and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.

Answer **all** questions.

Write your answers in the space provided in the question paper.

Shift	Laboratory

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

Give details of the practical shift and laboratory in the boxes provided.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

FOR EXAMINER'S USE **Score**	
1	**/21**
2	**/20**
3	**/14**
Total	**/55**

This document consists of **22** printed pages (including this cover page and **1** blank page).

List of Apparatus and Materials

Please note that items **1 – 3** are only available to you for the first 75 minutes of this examination.

PLACE ITEMS **2-3** BACK IN THE BASKET AFTER 75 MINUTES.

Item	**Apparatus / Reagents / Chemicals**	**Quantity per student**	**Time allocation**
1	Microscope with an eyepiece graticule	1 between 2	0 – 75 minutes
2	Prepared slide, labelled **S1**	1 between 2	
3	30 cm ruler (cm/mm), clear plastic	1	

Item	**Apparatus / Reagents / Chemicals**	**Quantity per student**
4	Pieces of fresh pondweed *(Cabomba* sp.), approximately 5 cm in length, in a 400 cm^3 beaker of 1% sodium hydrogencarbonate solution, labelled **P.**	5
5	50 cm^3 beaker, labelled **D**	50 cm^3
6	50 cm^3 syringe	1
7	25cm long glass capillary tube (1 mm diameter bore)	1
8	Rubber tubing	1
9	Bandage gauze (Dimensions: 10 x 10 cm)	10
10	Retort stand and clamp	1
11	Bench lamp	2
12	50 cm^3 beaker	1
13	Marker pen	1
14	Ethanol, labelled **E**	1
15	30 cm ruler (cm/mm)	1
16	Stopwatch	1
17	Safety glasses/goggles	1 pair
18	Paper towels	10
19	Scissors	1 pair
20	Blunt forceps	1
21	Sticky tape	at least 30 cm length

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Question 1

For Examiner's use

In this question you will investigate the effect of light intensity on the rate of photosynthesis.

(a) Sketch a fully-labelled graph to show the expected relationship between the rate of photosynthesis and light intensity, as light intensity increases.

Explain the shape of your graph. [4]

In your investigation, light intensity will be controlled using five filters, **F1, F2, F3, F4** and **F5.** To create the filters, wrap the corresponding number of bandage gauze as shown in Table 1.1. Note the percentage light intensity given by each filter when used with a standard light source.

Table 1.1

Filter*	Percentage light intensity / %	Number of gauze used
F1	100.0	0
F2	80.0	1
F3	60.0	2
F4	40.0	3
F5	20.0	4

* To create the filters, wrap the corresponding number of gauze together around the 50 cm^3 syringe, keeping them in place using sticky tape.

You are provided with:

- several pieces of pondweed *(Cabomba* sp.) in a beaker of 1% sodium hydrogencarbonate solution, labelled **P,** illuminated by a bench lamp
- several pieces of 10 x 10 cm bandage gauze to construct the light filters, **F1, F2, F3, F4** and **F5** (see Table 1.1)
- a 50 cm^3 syringe attached to a capillary tube by rubber tubing with an air-tight seal
- a 50 cm^3 beaker of 1% sodium hydrogencarbonate solution to which a small amount of detergent and food colouring has been added, labelled **D.**

Sodium hydrogencarbonate solution is a source of dissolved carbon dioxide.

*Read through steps **1** to **14** and prepare a table to record your results in **(b),** before starting the investigation.*

Proceed as follows:

Steps **1-9** outline the procedure to assemble the apparatus to measure the rate of photosynthesis of the pondweed. The apparatus is shown in Fig. 1.1 on page 5.

1 Use a retort stand and clamp to support the barrel of the 50 cm^3 syringe (with the plunger removed), so that the end of the capillary tube is over an empty beaker. Ensure that the positioning of the clamp is as high as possible on the syringe barrel so that it will not block light from reaching the pondweed.

2 Observe the pieces of pondweed in **P** to identify the piece that is releasing oxygen bubbles from its cut stem most rapidly. Request new material if there are no pieces that are actively bubbling.

3 Put the actively bubbling piece of pondweed into the barrel of the syringe and fill the syringe with the solution from **P,** as shown in Fig. 1.1. Use a finger to block the open end of the capillary tube while filling the barrel of the syringe. Tap the barrel to help release any bubbles trapped near the base of the syringe.

4 Remove your finger from the open end of the capillary tube and insert the plunger into the barrel of the syringe. Gently push the plunger down about 1 cm until some of the solution is forced out of the end of the capillary tube and there are no air bubbles in the capillary tube. Then gently pull the plunger up until there is about 1 cm of air from the bottom of the capillary tube.

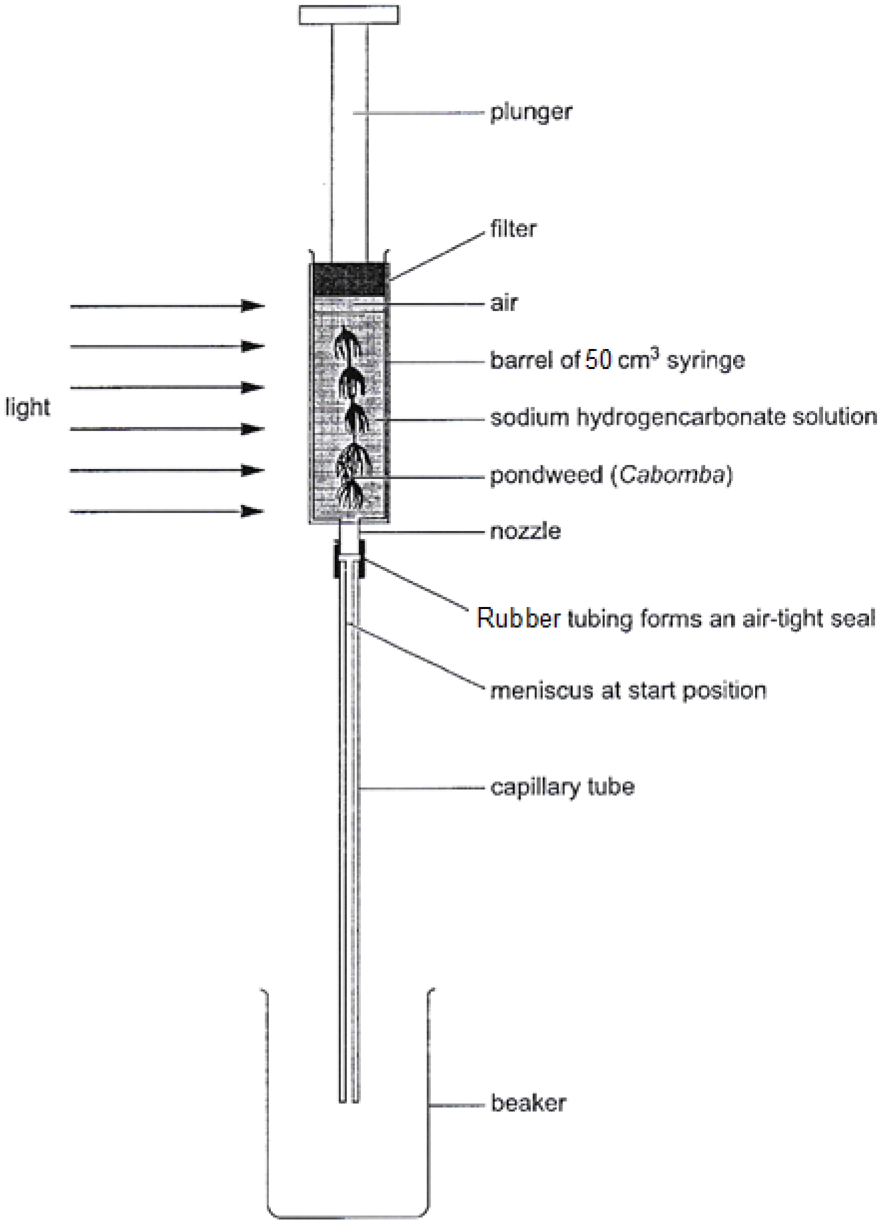

Fig. 1.1

For Examiner's use

5 Replace the empty beaker with the beaker containing solution **D** and adjust the clamp so that the open end of the capillary tube is below the surface of solution **D**. Carefully use the plunger to draw just enough of solution **D** into the capillary tube so that the air reaches, but does not pass, the nozzle of the syringe. This coats the inner surface of the capillary tube with detergent, which helps to ensure a smooth flow of liquid through the capillary tube during subsequent steps. Remove the beaker containing solution **D** and replace with the empty beaker.

6 Wipe the end of the capillary tube dry with a paper towel. Ensure that there is a clear meniscus a little below the plastic tubing, as shown in Fig 1.1. This is the start position. Mark the start position with a marker pen.

7 Position a lamp so that it will shine directly on to the syringe containing the pondweed. The lamp should be as close as possible to the pondweed, but no closer than 10 cm. The lamp should remain in this position throughout the investigation.

8 Switch on the lamp. From this point on, do **not** switch off the lamp. The pondweed should be maintained under constant illumination.

9 Carefully wrap filter **F1** around the barrel of the syringe and secure in place using sticky tape, on the side opposite to the light source. Leave the apparatus for two minutes to equilibrate.

10 If, after two minutes, the meniscus has moved down the capillary tube, continue from step **11.** If the meniscus is not moving, select a new piece of pondweed that is actively bubbling and re-set the apparatus by repeating steps **1** to **9**.

11 Use the plunger of the syringe to draw the meniscus back up to the mark for the starting position. Start a stopwatch and, after two minutes, mark the final position of the meniscus with a marker pen. Record the distance moved by the meniscus in two minutes, in the table prepared in **(b)**.

If the flow of liquid through the capillary tube becomes jerky when taking measurements, carefully push the plunger down until there are no air bubbles in the capillary tube. Then repeat steps **5** and **6** to reset the level of the meniscus to the start position.

12 Carefully remove filter **F1** and clean off the final position mark on the capillary tube (from step **11)** using ethanol, **E**.

13 Repeat step **9** with filter **F2** followed, after two minutes, by step **11.** Step **10** should not be repeated. After recording the result for filter **F2** in the table prepared in **(b),** remove filter **F2** and clean off the final position mark on the capillary tube using ethanol, **E**.

14 Repeat step **13** with filters **F3, F4,** and **F5,** in turn.

(b) Record your results at each light intensity in a suitable form in the space below. [5]

For Examiner's use

(c) Howard is required to conduct a similar experiment using acetate filters with varying optical densities. He proceeded to carry out steps **1** to **14** diligently. His results are shown in Table 1.2.

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Table 1.2

Acetate Filter	**Percentage light intensity / %**	**Distance travelled by meniscus / mm**
F1	90.0	35
F2	70.0	20
F3	50.0	13
F4	30.0	10
F5	10.0	8

(i) Use the grid on the next page to display Howard's results. [4]

(ii) Discuss what these results suggest about the relationship predicted in part **(a).** [2]

(d) Suggest the limiting factor for photosynthesis that is acting when filter **F5** is used. Explain your answer. [1]

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(e) Suggest how adding sodium hydrogencarbonate solution to the pondweed in the syringe increases the validity of the results. [2]

(f) One way to increase confidence in the conclusions of this investigation would be to repeat the experiment several times.

Describe **three** other modifications to the method that would increase confidence in the conclusions, and explain how these modifications would achieve this. [3]

[Total: 21]

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Question 2

During this question you will require access to a microscope and slide **S1**.

Fig. 2.1 is a photomicrograph of a stained transverse section through a plant stem.

The stem of this plant grows submerged in water and contains air spaces.

You are not expected to be familiar with this specimen.

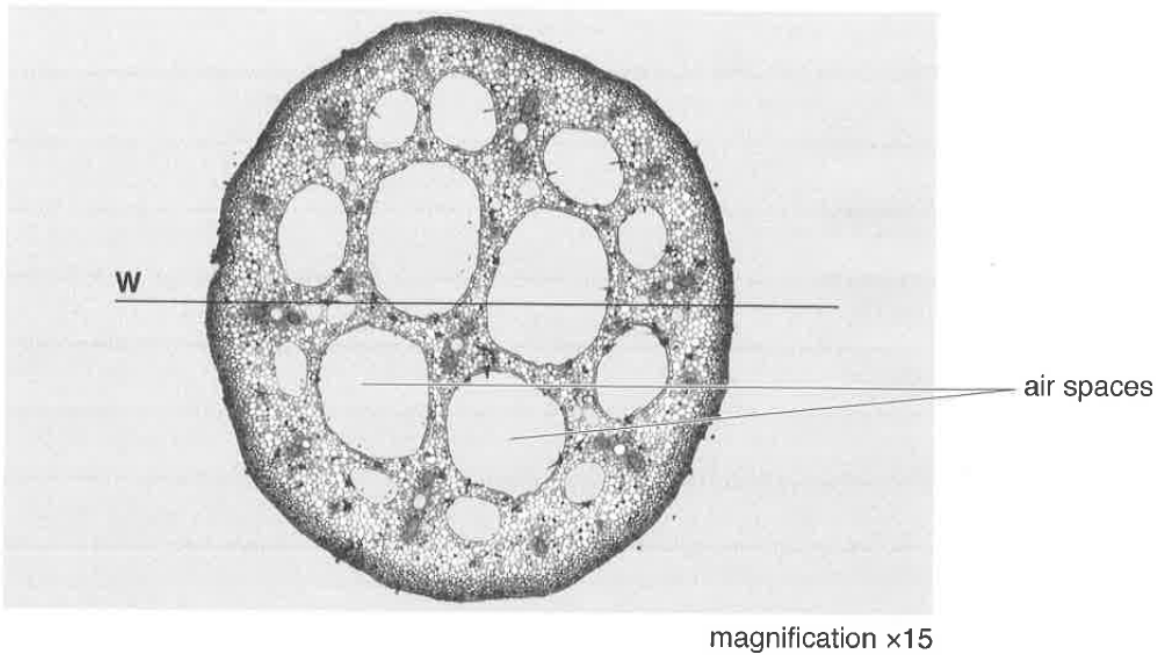

Fig. 2.1

Slide **S1** is a microscope slide of a stained transverse section through a stem from a **different** species of plant. This stem also grows submerged in water and contains air spaces.

(a) Use a suitable table to record one observable similarity and difference between the specimen in Fig. 2.1 and the specimen on slide **S1**. [2]

(b) (i) Calculate the actual radius of the stem at the position marked by line **W** in Fig. 2.1. You should show your working and use appropriate units. [3]

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Actual radius of stem = ____________________

You are required to estimate the radius of the stem on slide **S1**.

(ii) Put the clear plastic ruler on the stage of the microscope and view the scale lines on it using low power (x10 objective lens).

Estimate the diameter of the field of view to 1 decimal place of a mm. [1]

Diameter of field of view = ____________________

(iii) View the stem on slide **S1** using low power.

Estimate the fraction of the diameter of the field of view occupied by the radius of the stem on slide **S1**. [1]

Fraction of diameter of field of view = ____________________

(iv) Using your estimates from **(b)(ii)** and **(iii)**, calculate the radius of the stem on slide **S1**, using appropriate units. [1]

Radius of **S1** = ____________________

(v) Describe how to obtain a more accurate measurement of the radius of the stem on slide **S1**. State any additional pieces of apparatus that you might need. [3]

(c) (i) You are required to use a sharp pencil for drawings.

Use the space provided to draw a plan diagram of part of the stem on slide **S1**, as shown in the shaded area of Fig. 2.2. A plan diagram only shows the arrangement of the different types of tissues. Individual cells must **not** be drawn in plan diagrams.

Within this part of the stem there will be a number of air spaces.

You should only draw **three** of these air spaces.

Your drawing should show the correct shape and proportion of the tissues **and** three air spaces. [4]

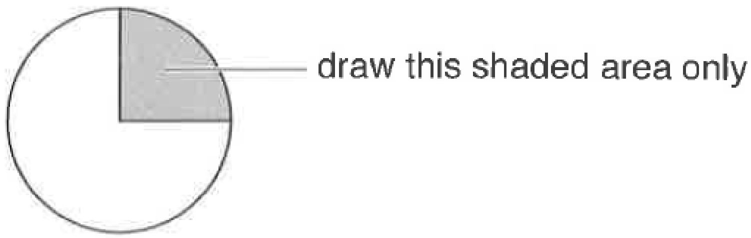

Fig. 2.2

For Examiner's use

(ii) Observe the cells that are found between air spaces in slide **S1**.

Select **one** group of **three** touching cells that are found between two air spaces.

Each cell of the group must touch at least one of the other cells.

Make a large drawing of this group of **three** cells. [4]

(d) Suggest **one** advantage of having air spaces in plant stems that grow submerged in water, as shown in Fig. 2.1 and slide **S1**. [1]

[Total: 20]

Question 3

For Examiner's use

All green plants photosynthesise in the light, taking in carbon dioxide and releasing oxygen. They also respire continuously, taking in oxygen and releasing carbon dioxide. The light intensity, at which photosynthesis and respiration occur at the same rate so that there is no net gas exchange, is called the compensation point.

Compensation points can be investigated using hydrogencarbonate indicator solution. This is harmless to living organisms but changes colour over a range of concentrations of carbon dioxide due to changes in pH, as shown in Table 3.1.

Table 3.1

Percentage concentration of carbon dioxide	Colour
0.04 (normal atmospheric)	red
falling below 0.04	turns purple
rising above 0.04	turns yellow

Monitoring the colour of hydrogencarbonate indicator solution in sealed vessels containing plant material can show whether carbon dioxide is being taken in or given out.

One factor that can affect the compensation point is whether a leaf is adapted for low light intensities (shade leaf) or high light intensities (sun leaf). Shade leaves would be expected to reach their compensation points at a lower light intensity than sun leaves.

Some plants produce both shade and sun leaves depending on where the leaves develop. For example, a single ivy plant can produce sun leaves at the top where they are in direct sunlight and produce shade leaves lower down where light intensity is reduced.

Using this information and your own knowledge, design an experiment to find the light intensity at which shade leaves and sun leaves from an ivy plant reach their compensation points.

Comparison of the results would then allow testing of the hypothesis that shade leaves reach their compensation points at a lower light intensity than sun leaves.

You must use:

- hydrogencarbonate indicator solution
- sun and shade leaves from ivy.

You may select from the following apparatus and use appropriate additional apparatus:

- normal laboratory glassware, e.g. test-tubes, boiling tubes, beakers, measuring cylinders, graduated pipettes, glass rods, etc.
- syringes
- timer, e.g. stopwatch
- bungs
- bench lamp with 60W bulb.

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it
- be illustrated by relevant diagram(s), if necessary, to show, for example, the arrangement of the apparatus used
- identify the independent and dependent variables
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and repeatable as possible
- include layout of results tables and graphs with clear headings and labels
- use the correct technical and scientific terms
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 14]

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